IT SKILLS TRAINING NEPAL

Gairidhara, Kathmandu, Nepal

DATA SCIENCE FOUNDATIONAL COURSE

Official Course Syllabus • 1.5 Months • Beginner to Intermediate

Duration	6 Weeks (1.5 Months)	**Mode**	In-person / Online
Total Hours	~72 Hours	**Schedule**	5 Days/Week, 2.5 Hrs/Day
Level	Beginner to Intermediate	**Language**	English / Nepali
Instructor	ISTN Expert Faculty	**Certificate**	Yes, Upon Completion

01 Course Overview

This foundational Data Science course is designed for students, fresh graduates, and working professionals who want to break into the world of data. Over 6 intensive weeks, students will build a strong foundation in Python programming, data analysis, statistics, machine learning, and real-world project deployment — guided by expert ISTN instructors.

Who Should Enroll?

- Students from any academic background (Science, Management, IT)
- Fresh graduates looking to enter the data field
- Working professionals seeking upskilling in data analytics
- Entrepreneurs and marketers wanting to leverage data insights

Prerequisites

- Basic computer skills (MS Office, internet browsing)
- No prior programming experience required
- Curiosity and willingness to learn

02 Learning Outcomes

By the end of this course, students will be able to:

- Write Python code for data manipulation and automation
- Work with real datasets using Pandas and NumPy
- Create compelling data visualizations using Matplotlib and Seaborn
- Understand and apply statistical concepts to analyze data

- Build and evaluate machine learning models using Scikit-learn
- Complete an end-to-end capstone data science project
- Present data-driven insights professionally

03 Weekly Curriculum

WEEK 1 Python Programming Fundamentals

Day	Topic	Key Learning Points
Day 1	**Introduction to Data Science**	What is DS, career paths, tools overview, environment setup (Anaconda, Jupyter)
Day 2	**Python Basics I**	Variables, data types, operators, input/output, string manipulation
Day 3	**Python Basics II**	Lists, tuples, dictionaries, sets, and their methods
Day 4	**Control Flow**	if/elif/else conditions, for/while loops, break/continue
Day 5	**Functions & Modules**	Defining functions, arguments, return values, importing modules

WEEK 2 Data Handling with Pandas & NumPy

Day	Topic	Key Learning Points
Day 1	**NumPy Essentials**	Arrays, array operations, broadcasting, reshaping, indexing
Day 2	**Pandas – Series & DataFrame**	Creating DataFrames, reading CSV/Excel, basic operations
Day 3	**Data Cleaning**	Handling missing values, duplicates, data type conversion
Day 4	**Data Transformation**	Filtering, sorting, groupby, pivot tables, merging datasets
Day 5	**Mini Project I**	Real-world dataset analysis: student/sales data EDA

WEEK 3 Data Visualization

Day	Topic	Key Learning Points
Day 1	**Matplotlib Basics**	Line charts, bar charts, scatter plots, customization

Day	Topic	Key Learning Points
Day 2	**Advanced Matplotlib**	Subplots, figure styling, annotations, saving plots
Day 3	**Seaborn**	Statistical plots: heatmaps, boxplots, pairplots, violin plots
Day 4	**Plotly / Dashboard Intro**	Interactive charts, introduction to Plotly Express
Day 5	**Storytelling with Data**	Choosing the right chart, design principles, visualization best practices

WEEK 4 Statistics & Probability

Day	Topic	Key Learning Points
Day 1	**Descriptive Statistics**	Mean, median, mode, variance, standard deviation, IQR
Day 2	**Probability Basics**	Events, probability rules, conditional probability, Bayes' theorem
Day 3	**Distributions**	Normal, binomial, Poisson distributions; CLT overview
Day 4	**Correlation & Hypothesis Testing**	Pearson/Spearman correlation, p-value, t-test, chi-square
Day 5	**Mini Project II**	Statistical analysis of a real dataset with full reporting

WEEK 5 Machine Learning Fundamentals

Day	Topic	Key Learning Points
Day 1	**Intro to Machine Learning**	Types of ML (supervised/unsupervised/reinforcement), ML workflow
Day 2	**Linear & Logistic Regression**	Simple/multiple linear regression, classification with logistic regression
Day 3	**Decision Trees & Random Forest**	Tree splits, overfitting, ensemble methods, feature importance
Day 4	**Model Evaluation**	Train/test split, cross-validation, accuracy, precision, recall, F1, confusion matrix
Day 5	**Clustering (Unsupervised)**	K-Means clustering, elbow method, silhouette score

WEEK 6 Capstone Project & Career Preparation

Day	Topic	Key Learning Points
Day 1	**Capstone Project Kickoff**	Problem selection, dataset sourcing, project planning, team/individual setup
Day 2	**Project Development I**	EDA, cleaning, feature engineering with instructor guidance
Day 3	**Project Development II**	Model building, evaluation, iteration and optimization
Day 4	**Portfolio & Career Skills**	GitHub portfolio setup, writing a data science resume, LinkedIn optimization
Day 5	**Final Presentation & Graduation**	Project presentation, Q&A, certificates, career guidance session

04 Tools & Technologies Covered

Python 3.x	**Jupyter Notebook**	**NumPy**	**Pandas**
Matplotlib	**Seaborn**	**Scikit-learn**	**Plotly**
Git & GitHub	**VS Code**	**Anaconda**	**Google Colab**

05 Assessment & Certification

Component	Weight	Timing
Attendance & Participation	**10%**	Ongoing
Mini Project I — Data Analysis	**20%**	End of Week 2
Mini Project II — Statistical Report	**20%**	End of Week 4
Quizzes (2 total)	**10%**	Week 2 & Week 4
Final Capstone Project	**30%**	Week 6
Final Presentation	**10%**	Day 5, Week 6

Students who score 60% or above and complete the capstone project will receive an ISTN Certificate of Completion in Data Science Fundamentals.

06 Enroll & Contact

- Location: IT Skills Training Nepal (ISTN), Gairidhara, Kathmandu
- Phone / WhatsApp: [Your Phone Number]

- Email: info@istn.edu.np
- Website: www.istn.edu.np
- Facebook: facebook.com/istnnepal

Early enrollment discount available. Batch size is limited — register now!

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